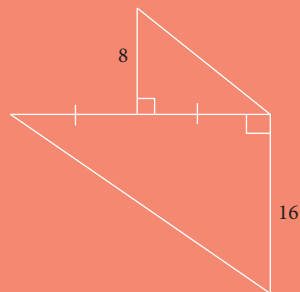


3

MIXED REVISION CHAPTERS 7 • 8 • 9

Multiple choice

- Convert 2 cm : 0.5 km to an equivalent ratio in simplest form.
 A 1 : 50 B 1 : 250 C 1 : 500
 D 1 : 25 000 E 1 : 250 000
- A survey was completed which collected the information about the heights of students within a cohort of Year 11 students. The type of data collected is best described as:
 A categorical and nominal data.
 B numerical and ordinal data.
 C categorical and ordinal data.
 D numerical and discrete data.
 E numerical and continuous data.
- The y -intercept of the line $3x - 2y + 8 = 0$ is:
 A -2 B $\frac{3}{2}$ C 3 D 4 E 8
- Which similarity test, if any, could be used to show that the two triangles are similar?
 A They are not similar. B SSS C SAS
 D AAA E RHS



- For the spelling test results given below; the range, mode, mean (rounded to one decimal place) and median respectively are:
 6, 5, 6, 7, 9, 8, 10, 7, 7, 7, 8, 6, 8, 9, 9, 7, 7
 A 5, 7, 7.4, 7 B 7.4, 7, 7, 5 C 5, 7, 7, 7.4
 D 1, 7, 7.4, 7 E 7, 7, 7, 7
- Calculate the gradient of the line connecting $(-5, 2)$ and $(3, 1)$.
 A -8 B $-\frac{1}{8}$ C $\frac{1}{8}$ D $\frac{1}{6}$ E 6
- A street directory has a scale of 1 : 16 000. What distance is represented by 35 mm?
 A 56 m B 560 m C 5.6 km D 56 km E 560 km
- For the spelling test results in Question 5, the IQR is:
 A 2 B 3.5 C 5 D 7 E 15

9 The equation of the graph at right is:

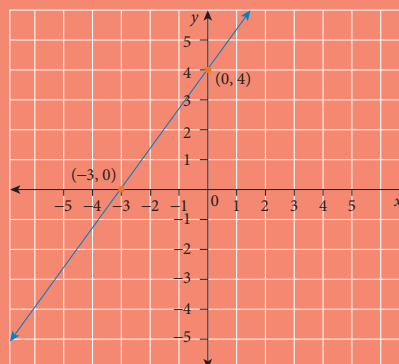
A $y = 4 - 3x$

B $y = 3 - 4x$

C $y = 4 + \frac{4}{3}x$

D $y = 4 - \frac{4}{3}x$

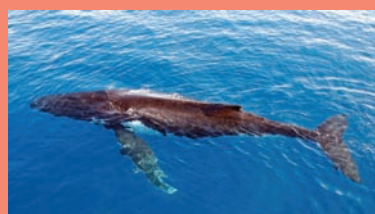
E $y = -3 + 4x$



Short answer questions

1 The picture of the whale has a scale of 1 : 300.

- If measuring from the tip of the nose to the tip of the tail, what is the length of the real whale?
- What is the tail width of the real whale?
- A fully grown male gray whale can grow up to 15 m long. How long would its picture be using the same scale?



Shutterstock.com/Jon Kratochvila

2 Motor vehicles were clocked, by police radar, travelling at the following speeds (in km/h).

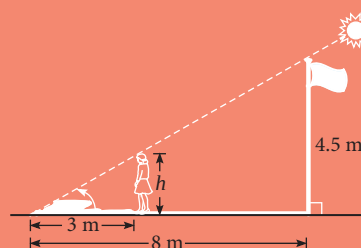
78 95 64 77 81 84 77 89 90 78 79 80 82 84 80 79 95 86 84 70
78 65 82 91 89 60 85 81 78 68 90 84 69 70 80 91 85 84 80 76

- Sort the data in a frequency table using classes of $60 - < 70$, $70 - < 80$, and so on.
- Use the frequency table to calculate an estimate for the mean speed rounded to one decimal place.
- State both the median class and modal class.

3 a What is the gradient and y -intercept of $2x - 4y = 12$?

b Sketch the line.

4 Ashleigh casts a 3 m shadow at the same time as a 4.5 m flagpole throws an 8 m shadow. Calculate h , Ashleigh's height in metres, to the nearest cm.



5 The following information represents the response times (in seconds) for emergency calls by 000 telephone operators:

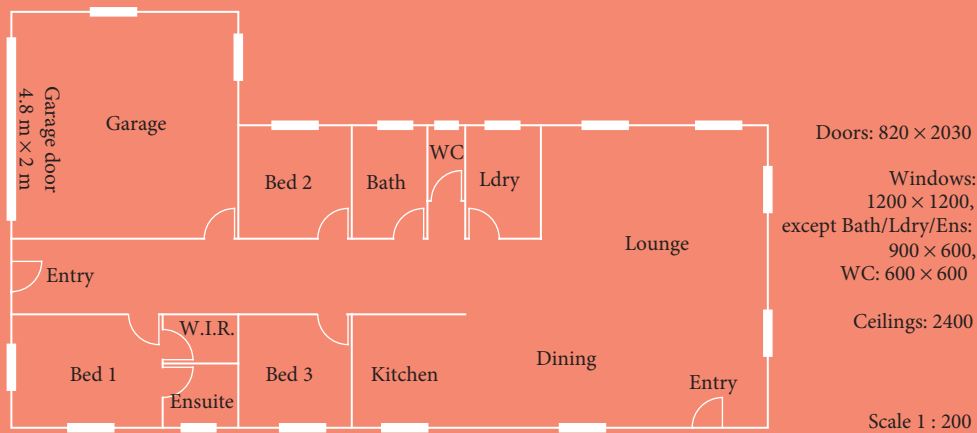
12 75 45 52 61 49 53 45 64 22 54 15 23 45 68 51 40
39 60 61 55 58 41 69 66 52 38 79 18 54 22 31 47

- Construct a stem-and-leaf plot to display the data.
- Calculate the five-number summary.
- Use your answers to part **b** verify if there are any outliers in the data.

6 Sketch $x = 4$ and $y = -2$ on the same number plane.

Application questions

- 1 One albatross with body length 135 cm had a wing span of 3 m. A similar albatross had a wing span of 2.5 m.
- a What was the body length of the smaller bird?
 - b What was the scale factor of the surface areas of their wings?
 - c What was the scale factor of their volumes?
 - d If the larger albatross weighed 11.88 kg, what would you expect the smaller bird to weigh?



- 2
- a Find the length of 3.66 m wide carpet needed to carpet the three bedrooms and the walk-in robe.
 - b Find the number of rolls of wallpaper needed for the walls of Bed 2, given that the chosen wallpaper comes in 50 cm wide rolls of length 10 m, with no pattern-matching required.
 - c Find the cost of painting the ceilings of the house and garage with two coats, given that the paint covers 11 m²/L and only comes in 4 L tins costing \$78.60 each. Fully explain your reasoning.

- 3 The ages, in years, of a sample of patients at a hospital are shown in the stem-and-leaf plot on the right.

Key: 1|2 means 12 years old

Key: 1|2 means 12 years old

- a Calculate the mean age of the patients.
- b Find the median age of the patients.
- c Is the mean or median more appropriate for describing the average age of the patients? Justify your reasoning.
- d Calculate the interquartile range of the patients' ages.
- e Represent this data set on a box-and-whisker plot.

Stem	Leaf
1	2 2 3 4 6
2	1 2
3	0 0 0 3
4	4 7 8
5	1 1
6	
7	5 7 8
8	1

- 4 The cost, C , of hiring a function room includes a set fee of \$250 and \$5 per head for the number of people attending. The table below shows the total cost \$ C for a different number of people, n , attending the function.

No of students (n)	20	30	40	50	100
Total cost (\$ C)	350	400	450	500	750

- Which variable is the dependent variable?
 - Explain how you can determine if the relationship shown is linear.
 - Calculate the gradient of the linear relationship.
 - What is the vertical intercept?
 - Write the linear function of C in terms of n .
 - What does the vertical intercept represent?
 - What would be the cost of the function if 135 people attended?
- 5 A plane is travelling at a constant speed. The graph below shows the distance travelled, d , after time, t , hours.
- How far will the plane travel in 2 hours?
 - How long will it take the plane to travel 4000 km?
 - Calculate the gradient of the linear relationship between d and t .
 - What is the vertical intercept?
 - Write the linear function of d in terms of t .
 - What does the vertical intercept represent?
 - What does the gradient represent?

